

Thesis Report

Spatial Tensor Simulation

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Abstract—TBD

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1. SPATIAL DATA

Spatial data can be easily generated using a simple grid. For the purposes of comparing how spatial smoothers behaves we will ganarte a small squire grid of 10 × 10. The following figure shows the grid used for this simulation.

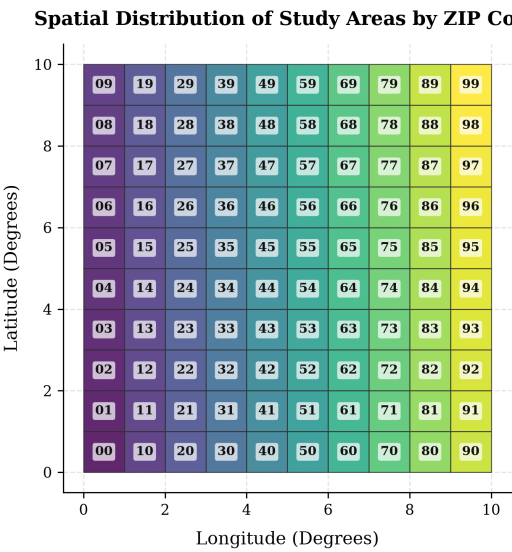


Figure 1. Simulated Zipcode Grid Map

As this not a single point in time, we look how this each spatial agent behaves thought space. This this we will generate a balance panel dataset [1]

This simulated will be generate an outcome variable that it distribute from a normal distributions were the data is infunced spatialy by its neoughrs. This can be summerized with the follwoign model

$$\begin{aligned} Y &\sim N(\alpha + \beta_1 X_1 + \beta_2 X_2 + \rho WY) \\ X_i &\sim N(2, 3), \quad \text{For } 1, 2, 3 \end{aligned} \tag{1}$$

REFERENCES

[1] G. Chamberlain, “Panel data”, *Handbook of econometrics*, vol. 2, pp. 1247–1318, 1984.

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